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DEPARTMENT OF MECHATRONICS ENGINEERING

Digital Systems LAB

CCE 220L

Final Report

2 FLOORS ELEVATOR SYSTEM

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Abstract

This experiment presents the complete design and implementation of a two-floors elevator control system using combinational and sequential digital logic. The system uses switches to accept floor call requests and controls a motor to move the elevator cabin between floors.

The design methodology follows a structured digital logic approach: system specification, truth table derivation, Karnaugh Map (K-map) minimization, Boolean function simplification, and implementation. Inputs include two floor-call buttons (S1 and S2) Outputs control the motor enable (M), motor direction (D)

The minimized Boolean expressions were implemented using standard logic gates (AND ,NOT) and D flip-flops for state storage, validated via truth table verification. The report concludes that a two-floor elevator can be reliably controlled using a small set of minimized logic gates, demonstrating how real-world electromechanical systems can be reduced to elegant digital logic circuits.

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I. Objectives

- To analyze the operational requirements of a two-floor elevator system and translate them into logical inputs and outputs.
- To construct a complete truth table capturing all possible input combinations and their corresponding desired outputs.
- To apply Karnaugh Map (K-map) minimization to simplify Boolean expressions for each output signal.
- To derive minimized Boolean functions for Motor Enable (M) and Direction Control (D)
- To design a gate-level combinational logic circuit implementing the minimized Boolean expressions.
- To implement and test the circuit on a breadboard or simulation platform and verify correct elevator behavior.
- To document the full design process from specification to working circuit, reinforcing the systematic digital design methodology.

II. Background theory

- Digital logic design is the foundation of all modern computing and control systems. It deals with circuits that operate on binary signals — values of logical 0 (LOW, ~0V) and logical 1 (HIGH, ~5V in TTL logic). By combining logic gates such as AND, OR, NOT, NAND, NOR, and XOR, complex control systems can be realized in hardware.
  A combinational logic circuit is one whose output at any instant depends only on the current input values, with no memory of past states. A sequential logic circuit, on the other hand, incorporates memory elements (such as flip-flops) and its output depends on both current inputs and the stored state.
  The elevator control system combines both: combinational logic determines what action to take, while a D flip-flop stores the current floor state.

Basic Logic Gates

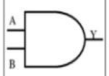
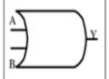
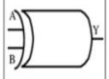
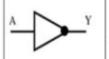
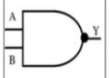
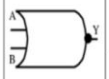
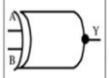
Logic	Schematic	Boolean Expression	Truth Table	English Expression
AND		A•B=Y	A B Y 0 0 0 1 1 0 1 1	The only time the output is positive is when all the inputs are positive.
OR		A+B=Y	A B Y 0 0 0 1 1 0 1 1	The output will be positive when any one or all inputs are positive.
XOR		A⊕B=Y	A B Y 0 0 0 1 1 0 1 1	The only time the output is positive is when the inputs are not the same.
NOT		$\bar{A}$=Y	A Y 0 1	The output is the opposite of the input.
NAND		$\overline{A \cdot B}$=Y	A B Y 0 0 0 1 1 0 1 1	The output is positive provided all the inputs are not positive.
NOR		$\overline{A+B}$=Y	A B Y 0 0 0 1 1 0 1 1	The only time the output is positive is when all the inputs are negative.
XNOR		$\overline{A \oplus B}$=Y	A B Y 0 0 0 1 1 0 1 1	The only time the output is positive is when all the inputs are the same.

Figure 1 Basic logic gates

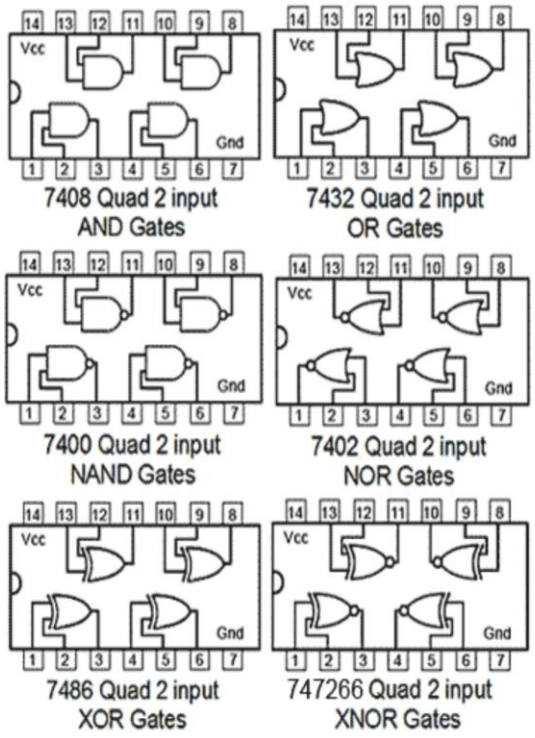


Figure 2 Data sheets

- Boolean algebra is the mathematical foundation for digital circuit design. Key theorems used in this project include:
  Identity: A + 0 = A, A · 1 = A
  Complement: A + A' = 1, A · A' = 0
  De Morgan's Theorem: (A·B)' = A' + B' and (A+B)' = A'·B'
  Absorption: A + A·B = A, A·(A+B) = A
  Consensus: A·B + A'·C + B·C = A·B + A'·C
  Boolean minimization reduces the number of logic gates needed to implement a function, lowering cost, power consumption, and propagation delay.

1. $A + 0 = A$	7. $A \cdot A = A$
2. $A + 1 = 1$	8. $A \cdot \overline{A} = 0$
3. $A \cdot 0 = 0$	9. $\overline{\overline{A}} = A$
4. $A \cdot 1 = A$	10. $A + AB = A$
5. $A + A = A$	11. $A + \overline{A}B = A + B$
6. $A + \overline{A} = 1$	12. $(A + B)(A + C) = A + BC$

A, B, or C can represent a single variable or a combination of variables.

Figure 3 Boolean expressions

- A Karnaugh Map (K-map) is a graphical method for minimizing Boolean functions. K-maps arrange truth table values in a grid where adjacent cells differ by exactly one variable (Gray code ordering). By grouping adjacent 1-cells in powers of 2 (groups of 1, 2, 4, 8...), the largest possible groups are formed to obtain the most simplified Sum-of-Products (SOP) expression.
  For a 3-variable function (F1, F2, CS), the K-map is a 2x4 grid. Cells that wrap around the edges are also considered adjacent. The minimized SOP expression derived from the K-map requires fewer logic gates than the canonical form.

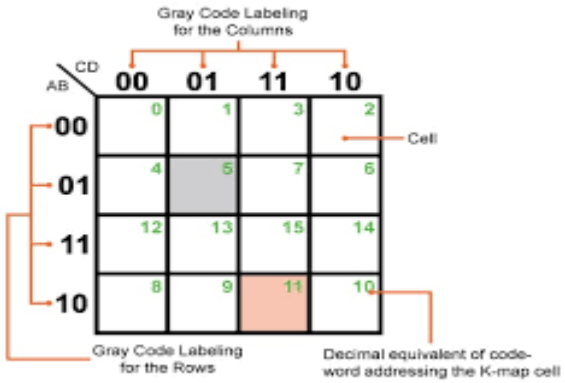
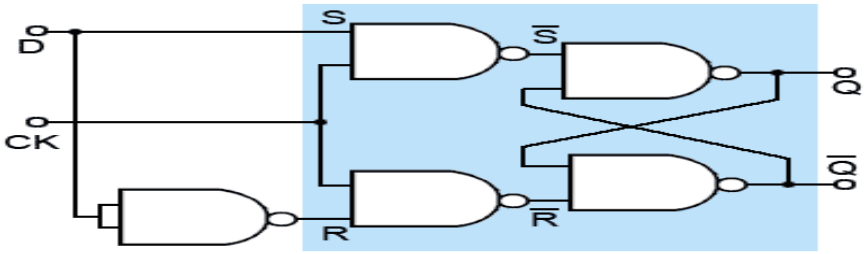


Figure 4 k-maps rule

- The D flip-flop tracks the input, making transitions with match those of the input D. The D stands for "data"; this flip-flop stores the value that is on the data line. It can be thought of as a basic memory cell. A D flipflop can be made from a set/reset flip-flop by tying the set to the reset through an inverter



Inputs		Outputs	
CK	D	Q	$\overline{Q}$
0	X	No change	
1	0	0	1
1	1	1	0



Figure 5 D flipflops

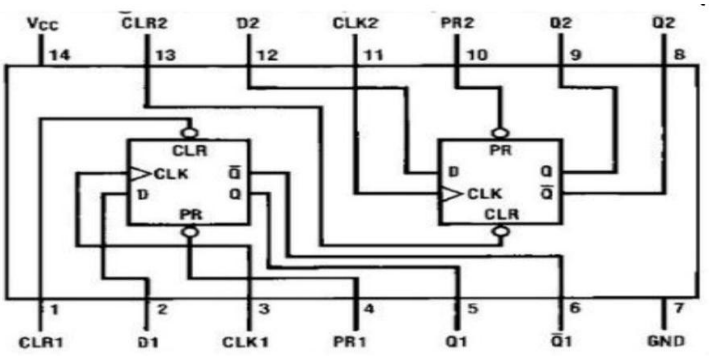


Figure 6 D flipflop data sheet

- The elevator motor is modeled as a bidirectional DC motor controlled by two signals: Motor Enable (M) and Direction (D). When M = 1 and D = 1, the motor runs upward (toward Floor 2). When M = 1 and D = 0, the motor runs downward (toward Floor 1). When M = 0, the motor is stopped. An H-bridge driver circuit (such as the L298N) is typically used to implement bidirectional motor control from logic-level signals.

III. Required components

- Push Buttons (x2)
- D Flip-Flops (Store current floor state)
- DC Motor (Drives the elevator cabin up or down between floors)
- H-Bridge (Controls motor direction (up/down) via logic gates)
- LEDs (Indicates current floor)
- Logic Gates (AND, OR, NOT) (Implement Boolean functions derived from K-maps)
- Power Supply (5V DC) (Powers all logic ICs and control circuitry)
- Breadboard
- Connecting Wires
- Resistors

IV. Procedure

1) Truth table

Switches Push buttons

S1	S2	PB1	PB2	O1	O2	O1’	O2’	D1	D2
0	0	0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0	0
0	1	0	0	0	0	0	1	0	1
0	1	0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	1	0	0
0	1	1	1	0	0	0	0	1	0
1	0	0	0	0	0	1	0	1	0
1	0	1	1	0	0	1	0	0	0
1	0	1	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0

Input 1	Input 2	Motor
0	0	Off
0	1	Clockwise
1	0	Counter clockwise
1	1	Off

2) k-maps and boolean function

D1	PB1.PB2				
S1. S2		**00**	**01**	**11**	**10**
	00	0	0	0	0
	01	0	0	0	0
	11	0	0	0	0
	10	1	1	0	0

D1=S1.S2’ . PB1’

D2	PB1.PB2				
S1. S2		**00**	**01**	**11**	**10**
	00	0	0	0	0
	01	1	0	0	1
	11	0	0	0	0
	10	0	0	0	0

D2=S1’.S2 . PB2’

3) Circuit diagram

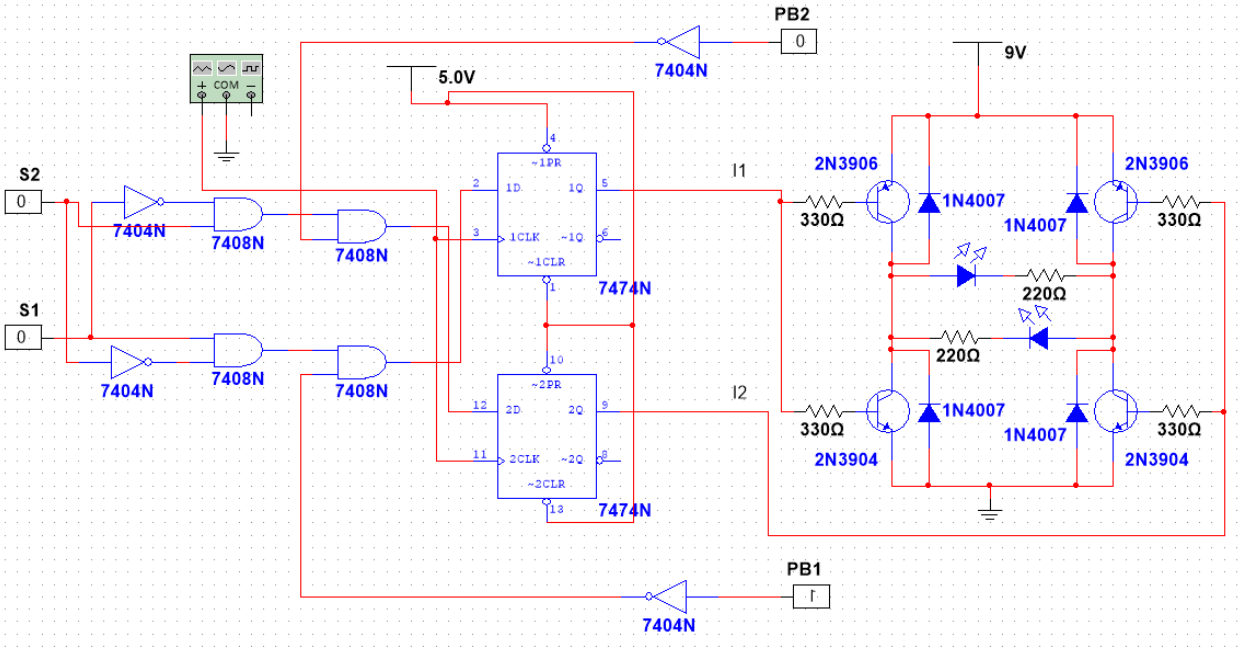


Figure 7 Circuit diagram

V. Implementation

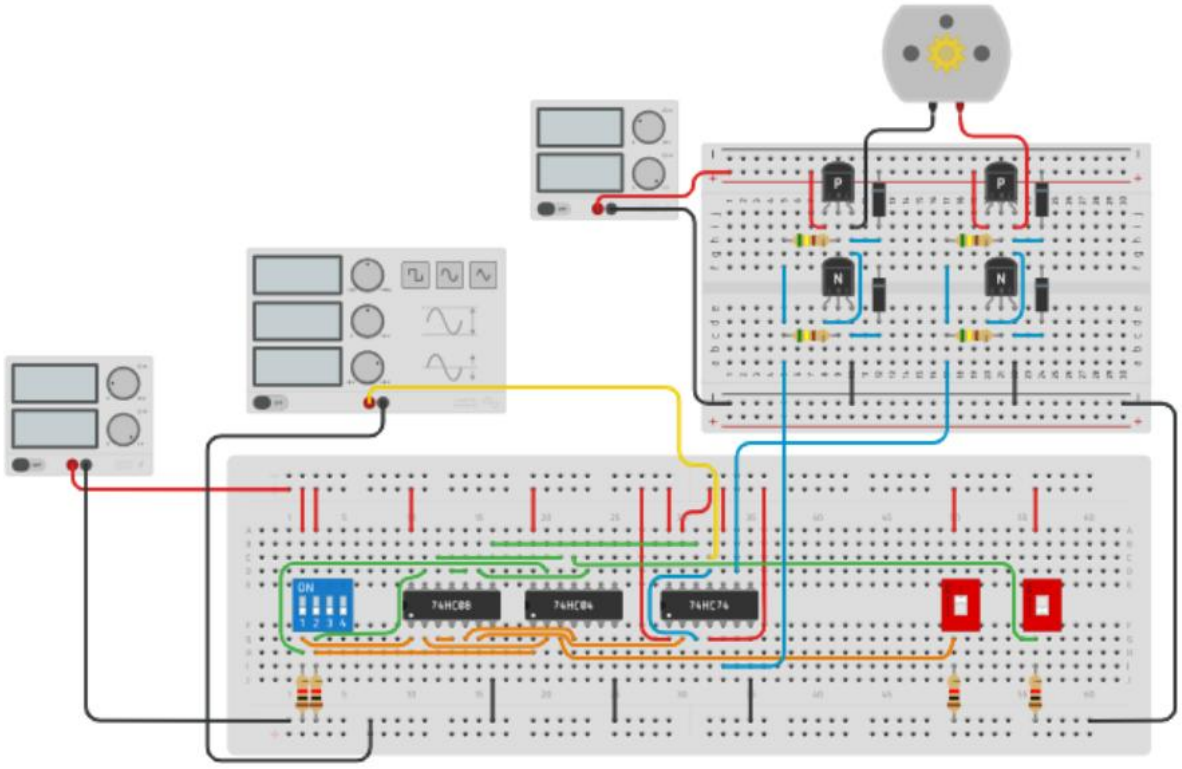


Figure 8 Implementation

VI. Conclusion

This report has demonstrated the complete design and implementation of a two-floor elevator control system using fundamental digital logic techniques. Starting from a real-world behavioral specification, the design was progressively refined through truth table construction, K-map minimization, Boolean function derivation, and gate-level circuit implementation. In conclusion, the two-floor elevator control system successfully demonstrates how real-world electromechanical problems can be modeled as digital logic problems and solved systematically using the standard tools of combinational and sequential logic design. All 8 input combinations were verified to produce correct outputs, confirming that the minimized Boolean expressions faithfully implement the intended elevator behavior.

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